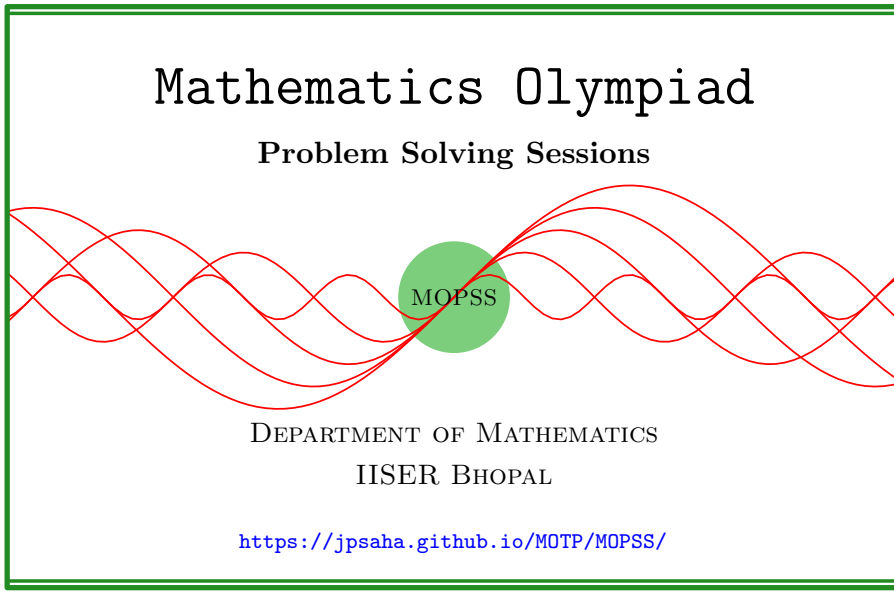
Quadratic polynomials



Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

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§1 Quadratic polynomials

Exercise 1 (All-Russian Mathematical Olympiad 2013 Grade 9 Day 1 P1, , by I. Bogdanov)**.** Given three distinct real numbers a, b, and c, show that at least two of the three following equations

$$(x-a)(x-b) = x-c,$$
$$(x-b)(x-c) = x-a,$$
$$(x-c)(x-a) = x-b$$

have real solutions.

> **Walkthrough** —
>
> **(a)**

Solution 1. Without loss of generality, let us consider the case that the equations

$$(x-a)(x-b) = x-c, (x-b)(x-c) = x-a$$

The content posted here and at this blog by Evan Chen are quite useful.

do not any real root. It follows that their discriminants are negative, that is,

$$(a+b+1)^2-4(ab+c), (b+c+1)^2-4(bc+a)$$

are negative. Note that

$$\begin{aligned}(a+b+1)^2-4(ab+c) &= (a-b)^2+1+2(a+b-2c)\\ &= (b-a+1)^2+4(a-c),\\ (b+c+1)^2-4(bc+a) &= (b-c)^2+1+2(b+c-2a)\\ &= (b-c+1)^2+4(c-a).\end{aligned}$$

Since a and c are distinct real numbers, one of $a-c$ and $c-a$ is positive. Thus, the discriminant of one of the equations

$$(x-a)(x-b)=x-c, (x-b)(x-c)=x-a$$

is non-negative, which is a contradiction. Therefore, at least two of the three equations have real solutions. ■

Solution 2. Let $f(x)$, $g(x)$, and $h(x)$ denote the polynomials defined by

$$\begin{aligned}f(x) &= (x-a)(x-b)-(x-c),\\ g(x) &= (x-b)(x-c)-(x-a),\\ h(x) &= (x-c)(x-a)-(x-b).\end{aligned}$$

Note that

$$\begin{aligned}f(x)+g(x) &= (x-b)(2x-a-c)-(2x-a-c)\\ &= (2x-a-c)(x-b-1).\end{aligned}$$

This implies that $f(x)+g(x)$ admits a real root. Hence, at least one of $f(x)$ and $g(x)$ is non-positive at some real number. If one of them admits a real root, then we are done. Otherwise, at least one of them is negative at some real number. Also note that if k is a real number satisfying $k \geq |a|+|b|+|c|+2$, then

$$\begin{aligned}f(k+|a|+|b|) &= (k+|a|+|b|+a)(k+|a|+|b|+b)-(k+|a|+|b|+c)\\ &\geq k^2-k-|a|-|b|-|c|\\ &\geq k-|a|-|b|-|c|\\ &> 0\end{aligned}$$

holds, and similarly,

$$g(k+|a|+|b|)>0$$

holds too. By the intermediate value theorem, at least one of $f(x)$ and $g(x)$ admits a real root. Interchanging the roles of f, g, and h, it follows that at least one of any two of the polynomials f, g, and h admits a real root. Therefore, at least two of the three equations have real solutions for x. ■

Exercise 2 (All-Russian Mathematical Olympiad 2007 Grade 8 Day 1 P1, , by O. Podlipsky)**.** Given reals numbers a, b, c, prove that at least one of three equations

$$x^2 + (a-b)x + (b-c) = 0,$$
$$x^2 + (b-c)x + (c-a) = 0,$$
$$x^2 + (c-a)x + (a-b) = 0$$

has a real root.

> **Walkthrough** —
> **(a)**

Solution 3. Note that the sum of the discriminants of the quadratic polynomials

$$x^2 + (a-b)x + (b-c), x^2 + (b-c)x + (c-a), x^2 + (c-a)x + (a-b)$$

are equal to

$$\begin{aligned}&(a-b)^2 - 4(b-c) + (b-c)^2 - 4(c-a) + (c-a)^2 - 4(a-b)\\ &= (a-b)^2 + (b-c)^2 + (c-a)^2,\end{aligned}$$

which is non-negative, and hence, at least one of the discriminants is non-negative. This implies that at least one of the three equations has a real root. ■

Solution 4. Consider the three quadratic polynomials defined by

$$f(x) = x^2 + (a-b)x + (b-c),$$
$$g(x) = x^2 + (b-c)x + (c-a),$$
$$h(x) = x^2 + (c-a)x + (a-b).$$

Note that

$$f(0) + g(0) + h(0) = 0,$$

which implies that at least one of $f(0)$, $g(0)$, $h(0)$ is non-positive. Without loss of generality, assume that $f(0) \leq 0$. If $f(0)$ is zero, then $x = 0$ is a root of $f(x)$. If $f(0) < 0$, then noting that

$$\begin{aligned}&f(2|a-b| + 2|b-c| + 1)\\ &= \frac{1}{2}(2|a-b| + 2|b-c| + 1)^2 + (a-b)(2|a-b| + 2|b-c| + 1)\\ &\quad + \frac{1}{2}(2|a-b| + 2|b-c| + 1)^2 + (b-c)\end{aligned}$$

$$
\begin{aligned}
&= \frac{1}{2}(2|a-b| + 2|b-c| + 1)\,(2|a-b| + 2|b-c| + 1 - 2(a-b)) \\
&\quad + \frac{1}{2}(2|a-b| + 2|b-c| + 1)^2 + (b-c) \\
&\geq 1
\end{aligned}
$$

holds, it follows from the intermediate value theorem that $f(x)$ has a real root. ■

Exercise 3 (All-Russian Mathematical Olympiad 2023 Grade 9 Day 1 P1, , by I. Bogdanov)**.** Let f, g be two monic quadratic polynomials such that each of the polynomials $f, g, f+g$ have two distinct real roots. Suppose that the difference of the roots of f is equal to the difference of the roots of g. Prove that the difference of the roots of $f+g$ is not bigger than the above common difference.

> **Walkthrough** —
>
> **(a)** Let the roots of f be α and $\alpha + d$, and the roots of g be β and $\beta + d$, where d denotes the common difference of the roots of f.
>
> **(b)** Determine the difference of the roots of $f + g$ in terms of α, β, d.

Solution 5. By the hypotheses, it follows that there are real numbers α, β, d such that $\alpha, \alpha + d$ are the roots of f, and $\beta, \beta + d$ are the roots of g. Thus, we have

$$
\begin{aligned}
f(x) &= (x-\alpha)(x-\alpha-d) \\
&= x^2 - (2\alpha + d)x + \alpha(\alpha + d) \\
g(x) &= (x-\beta)(x-\beta-d) \\
&= x^2 - (2\beta + d)x + \beta(\beta + d).
\end{aligned}
$$

Note that

$$
f(x) + g(x) = 2x^2 - 2(\alpha + \beta + d)x + (\alpha^2 + \beta^2 + d(\alpha + \beta)).
$$

The difference of the roots of $f + g$ is equal to

$$
\begin{aligned}
&(\alpha + \beta + d)^2 - 2(\alpha^2 + \beta^2 + d(\alpha + \beta)) \\
&= (\alpha + \beta)^2 + d^2 + 2d(\alpha + \beta) - 2(\alpha^2 + \beta^2) - 2d(\alpha + \beta) \\
&= d^2 - (\alpha - \beta)^2 \\
&\leq d^2.
\end{aligned}
$$

This completes the proof. ■

Exercise 4 (Moscow Mathematical Olympiad 2024 Grade 9 P1, by A. Doledenok, V. Retinsky)**.** Let a, b, c, d be nonzero real numbers satisfying

$$\frac{a}{b} + \frac{b}{a} = \frac{c}{d} + \frac{d}{c}.$$

Prove that the product of some two of the numbers a, b, c, d is equal to the product of the remaining two numbers.

Solution 6. Note that

$$\frac{a}{b} - \frac{c}{d} = \frac{d}{c} - \frac{b}{a}$$

holds, which yields

$$\frac{ad - bc}{bd} = \frac{ad - bc}{ac}.$$

If $ad - bc = 0$, then we are done. Otherwise, if $ad - bc$ is nonzero, then we obtain that

$$ac = bd.$$

Thus, in either case, the product of some two of the numbers a, b, c, d is equal to the product of the remaining two numbers. ■

Solution 7. Write

$$t = \frac{a}{b} + \frac{b}{a}.$$

Note that $x + \frac{1}{x} = t$ holds for $x = \frac{a}{b}$ and for $x = \frac{c}{d}$. Thus, the numbers $\frac{a}{b}$ and $\frac{c}{d}$ are the roots of the quadratic equation

$$x^2 - tx + 1 = 0.$$

If $\frac{a}{b} = \frac{c}{d}$, then $ad = bc$ holds. Otherwise, if $\frac{a}{b} \neq \frac{c}{d}$, then they are the two distinct roots of the quadratic equation, and by Vieta's formulas, we have

$$\frac{a}{b} \cdot \frac{c}{d} = 1,$$

which yields $ac = bd$. Thus, in either case, the product of some two of the numbers a, b, c, d is equal to the product of the remaining two numbers. ■

Exercise 5 (SMMC 2018 A2)**.** Ada and Byron play a game. First, Ada chooses a non-zero real number a and announces it. Then Byron chooses a non-zero real number b and announces it. Then Ada chooses a non-zero real number c and announces it. Finally, Byron chooses a quadratic polynomial whose three coefficients are a, b, c in some order.

(a) Suppose that Byron wins if the quadratic polynomial has a real root and Ada wins otherwise. Determine which player has a winning strategy.

(b) Suppose that Ada wins if the quadratic polynomial has a real root and Byron wins otherwise. Determine which player has a winning strategy.

Walkthrough — Without loss of generality, we may assume that $a = 1$.

(a) In (a), Byron has a winning strategy by choosing $b = -1$, and then considering the polynomial $ax^2 + cx + b$.

(b) In (b), Ada has a winning strategy by choosing c

$$\begin{cases} -4\sqrt{b} & \text{if } b > 0, \\ \frac{b^2}{8} & \text{if } b < 0. \end{cases}$$

In any of the above cases, one can check that the discriminant of each of the three possible polynomials is positive.

Solution 8. ■

Example 6 (Hungary MO 2001/02, Grades 11 and 12 — technical schools, P2)**.** Consider the following 2000 equations:

$$\begin{aligned} 1x^2 + 2 \cdot 2x + 3 &= 0 \\ 2x^2 + 2 \cdot 3x + 4 &= 0 \\ 3x^2 + 2 \cdot 4x + 5 &= 0 \\ &\vdots \\ 2000x^2 + 2 \cdot 2001x + 2002 &= 0. \end{aligned}$$

For each equation, consider the product of the sum of the real roots and the sum of their reciprocals (if it exists). What is the product of these products?

Solution 9. Note that a quadratic polynomial with nonzero constant term has nonzero roots, and hence the reciprocals of its roots exist. Moreover, if α, β denote the roots of a quadratic polynomial $ax^2 + bx + c$ with $ac \neq 0$, then $1/\alpha, 1/\beta$ are the roots of quadratic polynomial $cx^2 + bx + a$. Note that

$$\alpha + \beta = -\frac{b}{a}, \frac{1}{\alpha} + \frac{1}{\beta} = -\frac{b}{c}$$

Since

$$(2(n+1))^2 - 4n(n+2) = 4n^2 + 8n + 4 - 4n^2 - 8n = 4 \geq 0,$$

it follows that the given 2000 equations have real roots. The product of the products considered is equal to

$$\begin{aligned} &\frac{2^2 \cdot 2^2}{1 \cdot 3} \times \frac{2^2 \cdot 3^2}{2 \cdot 4} \times \frac{2^2 \cdot 4^2}{3 \cdot 5} \times \frac{2^2 \cdot 5^2}{4 \cdot 6} \times \cdots \times \frac{2^2 \cdot 2000^2}{1999 \cdot 2001} \times \frac{2^2 \cdot 2001^2}{2000 \cdot 2002} \\ &= \frac{4^{2000}}{1 \cdot 2} \frac{2^2 \cdot 3^2 \cdot \cdots \cdot 2001^2}{3^2 \cdot 4^2 \cdot \cdots \cdot 2000^2} \frac{1}{2001 \cdot 2002} \end{aligned}$$

$$= \frac{4^{2000} \cdot 2^2 \cdot 2001^2}{1 \cdot 2 \cdot 2001 \cdot 2002}$$
$$= \frac{2001}{1001} \times 2^{4000}.$$

■

Example 7 (Canada CMO 1971 P4)**.** Determine all real numbers a such that the two polynomials $x^2 + ax + 1$ and $x^2 + x + a$ have at least one root in common.

Solution 10. Let a be a real nuber such that the two polynomials $x^2 + ax + 1$ and $x^2 + x + a$ have at least one root in common. Let α denote a common root of these polynomials. The equations

$$\alpha^2 + a\alpha + 1 = 0, \alpha^2 + \alpha + a = 0$$

yield

$$(a - 1)\alpha = a - 1.$$

If $a \neq 1$, then $\alpha = 1$ and hence $a = -2$. This proves that $a = 1, -2$.

If $a = 1$, then the given polynomials have at least one root in common. If $a = -2$, then the given polynomials vanish at 1.

We conclude that $a = 1, -2$ are precisely all the real numbers such that the given polynomials have at least one common root. ■

Exercise 8 (RMO 2003 P6,). **.** Find all real numbers a for which the equation

$$x^2 + (a - 2)x + 1 = 3|x|$$

has exactly three distinct real solutions for x.

Solution 11. Let a be a real number such that $x^2 + (a - 2)x + 1 = 3|x|$ has exactly three distinct real solutions. Note that the equation

$$(x^2 + (a - 2)x + 1 - 3x)(x^2 + (a - 2)x + 1 + 3x) = 0$$

also has exactly three distinct real solutions. It follows that the discriminant of one of the polynomials $x^2 + (a - 2)x + 1 - 3x$, $x^2 + (a - 2)x + 1 + 3x$ vanishes, and the discriminant of the other is positive. The discriminants of these polynomials are

$$\begin{aligned}(a - 5)^2 - 4 &= a^2 - 10a + 21 \\ &= (a - 3)(a - 7), \\ (a + 1)^2 - 4 &= a^2 + 2a - 3 \\ &= (a + 3)(a - 1)\end{aligned}$$

respectively. It follows that a does not belong to $(-\infty,-3)\cup(-3,1)\cup(1,3)\cup(3,7)\cup(7,\infty)$, or equivalently, a belongs to $\{-3,1,3,7\}$.

Let us determine whether the given equation has three distinct real roots if a lies in $\{-3,1,3,7\}$. Let us assume that a lies in $\{-3,1,3,7\}$. Note that then one of $(a-3)(a-7),(a+3)(a-1)$ vanishes and another is positive, and consequently, one of the polynomials

$$x^2+(a-2)x+1-3x, x^2+(a-2)x+1+3x$$

have distinct real roots, and the other has a double root, which is a real number. Observe that if $x^2+(a-2)x+1-3x$ has a double root, then that root is equal to $-\frac{a-5}{2}$. Note that

$$\begin{aligned}\left(\frac{a-5}{2}\right)^2-(a+1)\left(\frac{a-5}{2}\right)+1&=\frac{1}{4}\left(a^2-10a+25-2a^2+8a+10+4\right)\\&=\frac{1}{4}\left(-a^2-2a+39\right),\end{aligned}$$

which has a negative discriminant. This shows that if $x^2+(a-2)x+1-3x$ has a real double root, then that cannot be a zero of $x^2+(a-2)x+1+3x$. Also note that

$$\begin{aligned}\left(\frac{a+1}{2}\right)^2-(a-5)\frac{a+1}{2}+1&=\frac{1}{4}\left(a^2+2a+1-2a^2+8a+10+4\right)\\&=\frac{1}{4}\left(-a^2+10a+15\right),\end{aligned}$$

whose roots are not integers. Using that a is an integer, it follows that if $x^2+(a-2)x+1+3x$ has a real double root, then that cannot be a root of $x^2+(a-2)x+1-3x$. We conclude that if a lies in $\{-3,1,3,7\}$, then

$$(x^2+(a-2)x+1-3x)(x^2+(a-2)x+1+3x)=0$$

has exactly three distinct real solutions, or equivalently, the equation

$$x^2+(a-2)x+1=3|x|$$

has exactly three distinct real solutions for x.

So the required real numbers are $a=-3,1,3,7$. ■

Example 9 (All-Russian MO 2007 Grade 8 P1)**.** If a,b,c are real numbers, show that at least one of the equations

$$\begin{aligned}x^2+(a-b)x+(b-c)&=0,\\x^2+(b-c)x+(c-a)&=0,\\x^2+(c-a)x+(a-b)&=0\end{aligned}$$

has a real solution.

Solution 12. The sum of the discriminants of the above quadratic polynomials is

$$\begin{aligned}&(a-b)^2-4(b-c)+(b-c)^2-4(c-a)+(c-a)^2-4(a-b)\\&=(a-b)^2+(b-c)^2+(c-a)^2,\end{aligned}$$

which is positive if not all of a, b, c are equal. Consequently, if not all of the three real numbers a, b, c are equal, then at least one of the quadratic polynomials

$$x^2+(a-b)x+(b-c), x^2+(b-c)x+(c-a), x^2+(c-a)x+(a-b)$$

has positive discriminant, and hence admits real solutions. Moreover, if all of a, b, c are equal, at least one (in fact, all) of the above polynomials admits a real root. ■

Exercise 10 (RMO 2007 P3,). Find all pairs (a, b) of real numbers such that whenever α is a root of $x^2+ax+b=0$, α^2-2 is also a root of the equation.

Solution 13. Let a, b be real numbers such that for any root α of $x^2+ax+b=0$, α^2-2 is also a root. Denote the roots of x^2+ax+b by α, β. There are the following possibilities.

(1) $\alpha^2-2=\alpha$, $\beta^2-2=\beta$,

(2) $\alpha^2-2=\beta$, $\beta^2-2=\alpha$,

(3) $\alpha^2-2=\beta^2-2=\alpha$,

(4) $\alpha^2-2=\beta^2-2=\beta$.

If $\alpha=\beta$, then these four cases are equivalent to

$$\alpha^2-2=\beta^2-2=\alpha=\beta,$$

which shows that α is equal to 2 or -1, and hence (a, b) is equal to $(-4, 4)$ or $(2, 1)$.

It remains to consider the case that $\alpha\neq\beta$, which we assume from now on.

In Case (1), α, β satisfy the equation $X^2-X-2=0$. So (α, β) is equal to $(2, -1)$ or $(-1, 2)$, and hence (a, b) is equal to $(-1, -2)$.

In Case (2), we have $\alpha^2-\beta^2=\beta-\alpha$, which gives $\alpha+\beta=-1$ (since $\alpha\neq\beta$). So

$$\alpha\beta=\frac{1}{2}(\alpha+\beta)^2-\frac{1}{2}(\alpha^2+\beta^2)=\frac{1}{2}(\alpha+\beta)^2-\frac{1}{2}(\alpha+\beta+4)=\frac{1}{2}-\frac{3}{2}=-1.$$

This shows that α, β are roots of the quadratic polynomial x^2+x-1, and hence, (a, b) is equal to $(1, -1)$.

In Case (3), note that α is equal to 2 or -1. Using $\beta^2 = 2 + \alpha$ and $\alpha \neq \beta$, it follows that (α, β) is equal to $(2, -2)$ or $(-1, 1)$, and hence (a, b) is equal to $(0, -4)$ or $(0, -1)$.

Similarly, in Case (4), (α, β) is equal to $(-2, 2)$ or $(-1, 1)$, which shows (a, b) is equal to $(0, -4)$ or $(0, -1)$.

So (a, b) is equal to one of $(-4, 4), (2, 1), (-1, -2), (1, -1), (0, -4), (0, -1)$.

Moreover, if (a, b) is equal to any of these six pairs, then it can be checked that for any root α of $x^2 + ax + b = 0$, $\alpha^2 - 2$ is also a root.

We conclude that all the required pairs are $(-4, 4)$, $(2, 1)$, $(-1, -2)$, $(1, -1)$, $(0, -4)$, $(0, -1)$. ■

Exercise 11 (RMO 2010 P2,). Let

$$P_1(x) = ax^2 - bx - c, P_2(x) = bx^2 - cx - a, P_3(x) = cx^2 - ax - b$$

be three quadratic polynomials where a, b, c are nonzero real numbers. Suppose there exists a real number α such that $P_1(\alpha) = P_2(\alpha) = P_3(\alpha)$. Prove that $a = b = c$.

Solution 14. Since $P_1(\alpha), P_2(\alpha)$ are equal, we get

$$(a - b)\alpha^2 = (b - c)\alpha + (c - a),$$

which gives

$$(a - b)(\alpha^2 + 1) = (b - c)(\alpha - 1).$$

Similarly, using $P_2(\alpha) = P_3(\alpha)$, we obtain

$$(b - c)(\alpha^2 + 1) = (c - a)(\alpha - 1).$$

If $\alpha = 1$, then it follows that $a = b = c$. Henceforth, let us assume that $\alpha \neq 1$. Then the above yields

$$(a - b)(c - a) = (b - c)^2.$$

Using a similar argument as above, it follows that

$$(b - c)(a - b) = (c - a)^2, (c - a)(b - c) = (a - b)^2.$$

Adding these equations, we obtain

$$a^2 + b^2 + c^2 = ab + bc + ca.$$

Since a, b, c are real, it follows that $a = b = c$. This completes the proof. ■

Exercise 12 (RMO 2012f P1,). Find nonzero real numbers a, b such that $x^2 + ax + b, x^2 + x + ab, ax^2 + x + b$ are three distinct polynomials with a common root.

Solution 15. Let a, b be real numbers such that $x^2+ax+b, x^2+x+ab, ax^2+x+b$ are three distinct polynomials with a common root $\alpha \in \mathbb{C}$. We obtain

$$\alpha^2 + a\alpha + b = \alpha^2 + \alpha + ab = a\alpha^2 + \alpha + b = 0,$$

which gives $a\alpha + b = \alpha + ab$, that is, $(a-1)(\alpha - b) = 0$. Since $x^2 + ax + b, x^2 + x + ab, ax^2 + x + b$ are distinct, it follows that $a \neq 1$. This shows that $\alpha = b$. Since the polynomials $x^2 + x + ab, ax^2 + x + b$ vanish at $x = \alpha = b$, we obtain

$$b(a + b + 1) = b(ab + 2) = 0.$$

Using b is nonzero, we get $a + b + 1 = ab + 2 = 0$. Note that $1 + ab - a - b = 0$. Since $a \neq 1$, we obtain $b = 1$, which combined with $ab + 2 = 0$ implies that $a = -2$.

Also note that for $a = -2, b = 1$, the given polynomials are equal to

$$x^2 - 2x + 1, x^2 + x - 2, -2x^2 + x + 1,$$

which are all distinct and they vanish at $x = 1$.

We conclude that precisely for $(a, b) = (-2, 1)$, the given polynomials are all distinct and have a common root. ■

Exercise 13 (RMO 2015d P2,). Let $P(x) = x^2 + ax + b$ be a quadratic polynomial with real coefficients. Suppose there are real numbers s is not equal to t such that $P(s) = t$ and $P(t) = s$. Prove that $b - st$ is a root of the equation $x^2 + ax + b - st = 0$.

Solution 16. We have

$$s^2 + as + b = t, t^2 + at + b = s.$$

Taking their difference, we obtain $(s - t)(s + t + a + 1) = 0$, which gives $s + t + a + 1 = 0$ since $s \neq t$. Using the above, we obtain

$$s(s^2 + as + b) - t(t^2 + at + b) = 0,$$

or equivalently,

$$(s - t)(b + a(s + t) + s^2 + st + t^2) = 0.$$

Combining the above with $s + t + a + 1 = 0$ and $s \neq t$, we obtain

$$b - (s + t) - st = 0.$$

Note that

$$\begin{aligned}(b - st)^2 + a(b - st) + b - st &= (b - st)(b - st + a + 1)\\ &= (b - st)(s + t + a + 1)\\ &= 0.\end{aligned}$$

This completes the proof. ■

Exercise 14 (RMO 2015a P2,). Let $P(x) = x^2 + ax + b$ be a quadratic polynomial where a, b are real numbers. Suppose $\langle P(-1)^2, P(0)^2, P(1)^2 \rangle$ is an AP of positive integers. Prove that a, b are integers.

Solution 17. Note that $P(-1)^2, P(0)^2, P(1)^2$ are equal to

$$1 + a^2 + b^2 - 2a + 2b - 2ab, b^2, 1 + a^2 + b^2 + 2a + 2b + 2ab$$

respectively. Since they form an arithmetic progression, we obtain

$$1 + a^2 + b^2 - 2a + 2b - 2ab + 1 + a^2 + b^2 + 2a + 2b + 2ab = 2b^2,$$

or equivalently, $a^2 + 2b + 1 = 0$. It follows that

$$b^2 - 2a - 2ab, b^2, b^2 + 2a + 2ab$$

form an arithmetic progression of positive integers. Note that

$$\begin{aligned}(2a + 2ab)^2 &= 4a^2(b + 1)^2 \\ &= -4(2b + 1)(b + 1)^2 \\ &= -4(2b^3 + 5b^2 + 4b + 1) \\ &= -4((2b^2 + 4)b + 5b^2 + 1).\end{aligned}$$

Since b^2 is an integer, it follows that b is rational number. Since b is rational and b^2 is an integer, it follows that b is an integer. Using $a^2 + 2b + 1 = 0$, it follows that a^2 is an integer. Moreover, if $b = -1$, then a is an integer. If $b \neq -1$, then using that $2a + 2ab$ is an integer, we obtain a is rational. Since a^2 is an integer and a is rational, it follows that a is an integer. This completes the proof. ■

Exercise 15 (RMO 2015b P2,). Let $P(x) = x^2 + ax + b$ be a quadratic polynomial where a is real and $b \neq 2$, is rational. Suppose $P(0)^2, P(1)^2, P(2)^2$ are integers, prove that a and b are integers.

Solution 18. Since b is rational and $P(0)^2 = b^2$ is an integer, it follows that b is an integer. Note that

$$\begin{aligned}P(1)^2 &= (1 + a + b)^2 \\ &= 1 + a^2 + b^2 + 2a + 2b + 2ab, \\ P(2)^2 &= (4 + 2a + b)^2 \\ &= 16 + 4a^2 + b^2 + 16a + 8b + 4ab.\end{aligned}$$

Since b is an integer, the given conditions imply that $a^2+2a+2ab, 4a^2+16a+4ab$ are integers. This shows that

$$4a^2 + 16a + 4ab - 2(a^2 + 2a + 2ab) = 2a^2 + 12a,$$

$$4a^2 + 16a + 4ab - 4(a^2 + 2a + 2ab) = 8a - 4ab$$

are integers. Since $b \neq 2$ and b is an integer, it follows that a is a rational number. Combining this with the fact that $2a^2 + 12a$ is rational, it follows that a is equal to $\frac{n}{2}$ for some integer n. Indeed, write $a = \frac{x}{y}$ where x, y are integers with $y \geq 1$ and $\gcd(x, y) = 1$. Note that $2\frac{x^2}{y} + 12x$ is an integer. Since x and y are relatively prime, this implies that y divides 2. Consequently, a is equal to $\frac{n}{2}$ for some integer n. Using that $2a^2 + 12a$ is an integer, we get that $\frac{n^2}{2}$ is also an integer. This shows that n is even, and hence a is an integer. This completes the proof. ■

Exercise 16 (RMO 2015e P2,). Let $P_1(x) = x^2 + a_1x + b_1$ and $P_2(x) = x^2 + a_2x + b_2$ be two quadratic polynomials with integer coefficients. Suppose $a_1 \neq a_2$ and there exist integers $m \neq n$ such that $P_1(m) = P_2(n), P_2(m) = P_1(n)$. Prove that $a_1 - a_2$ is even.

Solution 19. Using $P_1(m) = P_2(n)$, we get

$$m^2 + a_1m + b_1 = n^2 + a_2n + b_2,$$

that is,

$$(m^2 - n^2) + (a_1m - a_2n) + b_1 - b_2 = 0.$$

Similarly, using $P_1(n) = P_2(m)$, we get

$$(n^2 - m^2) + (a_1n - a_2m) + b_1 - b_2 = 0.$$

This yields

$$2(m^2 - n^2) + (a_1 + a_2)(m - n) = 0.$$

Since $m \neq n$, we get $2(m + n) + a_1 + a_2 = 0$. It follows that $a_1 + a_2$ is even, and hence, so is $a_1 + a_2 - 2a_2 = a_1 - a_2$. ■

Exercise 17 (RMO 2013c P4,). A polynomial is called a *Fermat polynomial* if it can be written as the sum of squares of two polynomials with integer coefficients. Suppose that $f(x)$ is a Fermat polynomial such that $f(0) = 1000$. Prove that $f(x) + 2x$ is not a Fermat polynomial.

> **Summary** — Assume that $f(x) + 2x$ is a Fermat polynomial. Write each of the polynomials $f(x), f(x) + 2x$ as a sum of the squares of two polynomials with integer coefficients. Reducing modulo 4, conclude that the constant terms of these four polynomials are even. Next, reduce modulo x^2, and compare the coefficients of x to obtain a contradiction.

Solution 20. On the contrary, let us assume that $f(x) + 2x$ is a Fermat polynomial.

Since $f(x)$ is a Fermat polynomial, it follows that there exist polynomials $P(x)$ and $Q(x)$ with integer coefficients such that $f(x) = P(x)^2 + Q(x)^2$. This gives $P(0)^2 + Q(0)^2 = f(0) = 1000$. Since a perfect square leaves one of $0, 1$ as a remainder upon division by 4, it follows that $P(0), Q(0)$ are even. Consequently, the coefficient of x in the polynomial $f(x) = P(x)^2 + Q(x)^2$ is a multiple of 4.

Since $f(x) + 2x$ is a Fermat polynomial, there exist polynomialss $R(x)$ and $S(x)$ with integer coefficients such that $f(x) + 2x = R(x)^2 + S(x)^2$. Using a similar argugemt as above, it follows that the coefficient of x in the polynomial $f(x) + 2x = R(x)^2 + S(x)^2$ is a multiple of 4.

Note that

$$2x = P(x)^2 + Q(x)^2 - R(x)^2 - S(x)^2,$$

and the coefficient of x in $P(x)^2 + Q(x)^2 - R(x)^2 - S(x)^2$ is a multiple of 4. This contradicts the assumption that $f(x) + 2x$ is a Fermat polynomial.

This proves that $f(x) + 2x$ is not a Fermat polynomial. ■

Exercise 18 (RMO 2023a P3,). Let $f(x)$ be a polynomial with real coefficients of degree 2. Suppose that for some pairwise distinct **nonzero** real numbers , a, b, c we have:

$$f(a) = bc, f(b) = ac, f(c) = ab.$$

Dertermine $f(a + b + c)$ in terms of a, b, c.

Solution 21. Note that cubic polynomial $xf(x) - abc$ has three roots a, b, c, and hence

$$xf(x) - abc = \lambda(x - a)(x - b)(x - c)$$

holds for some real number λ. Substituting $x = 0$ in the above, and using that a, b, c are nonzero, it follows that $\lambda = 1$. This gives $f(x) = x^2 - (a + b + c)x + ab + bc + ca$. We obtain that

$$f(a + b + c) = ab + bc + ca.$$

■

Exercise 19 (RMO 2024a P2,). Show that there do not exist non-zero real numbers a, b, c such that the following statements hold simultaneously:

1. the equation $ax^2 + bx + c = 0$ has two distinct roots x_1, x_2;
2. the equation $bx^2 + cx + a = 0$ has two distinct roots x_2, x_3;
3. the equation $cx^2 + ax + b = 0$ has two distinct roots x_3, x_1.

(Note that x_1, x_2, x_3 may be real or complex numbers.)

Solution 22. On the contrary, let us assume that there exist nonzero real numbers a, b, c such that the given condition holds. Note that

$$ax_2^2 + bx_2 + c = 0, bx_2^2 + cx_2 + a = 0$$

hold. Eliminating x_2^2, we obtain

$$(b^2 - ca)x_2 = a^2 - bc.$$

Also note that

$$(a^2 - bc)x_2^2 = (c^2 - ab)x_2$$

holds. Since $c \neq 0$, it follows that $x_2 \neq 0$, and hence

$$(a^2 - bc)x_2 = c^2 - ab.$$

Combining the above, we obtain

$$(a^2 - bc)^2 = (b^2 - ca)(c^2 - ab),$$

which can be simplified to

$$a(a^3 + b^3 + c^3 - 3abc) = 0.$$

Since a is nonzero, it implies that $a^3 + b^3 + c^3 = 3abc$, and hence $a + b + c = 0$ or $a = b = c$.

Note that if a, b, c satisfy $a = b = c$, then each of x_1, x_2, x_3 is equal to one of ω, ω^2, and hence, two of x_1, x_2, x_3 are equal, which is impossible by our assumption.

Suppose a, b, c satisfy $a + b + c = 0$. Note that if some two of a, b, c are equal, then one of the polynomials $ax^2 + bx + c, bx^2 + cx + a, cx^2 + ax + b$ does not have distinct roots. Note that if the roots of $ax^2 + bx + c$ are equal, then $1 = -b/a - 1$ holds, which implies that $b = -2a$, and hence $a = c$, which is impossible. Hence, the roots of $ax^2 + bx + c$ are distinct. By similar arguments, it follows that each of $bx^2 + cx + a, cx^2 + ax + b$ possesses distinct roots. If the common root of $ax^2 + bx + c, bx^2 + cx + a$ is equal to 1, then considering the common root of $bx^2 + cx + a, cx^2 + ax + b$, we obtain $c/b = a/c$, which gives $c^2 = ab$, and this yields $(a + b)^2 = ab$, which is impossible since a, b are nonzero real numbers. If the common root of $ax^2 + bx + c, bx^2 + cx + a$ is not equal to 1, then we obtain $b/a = c/b$, which implies that $(c + a)^2 = b^2 = ac$, which is impossible since a, c are nonzero real numbers.

This completes the proof. ■

References

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